

F1Tenth Odometry-Calibrated Anti-Skid System

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Abstract—The F1Tenth Odometry-Calibrated Anti-Skid System is an automatic odometry calibrator and online slip compensator for F1Tenth autonomous race cars. Manual tuning of vehicle odometry is a cumbersome task: a car tuned in simulation requires further tuning for real-world operation and for subsequent changes to track parameters. In addition, a car tuned for a high-traction surface tends to slip or skid when operated at high speeds on a low-traction surface. The goal of this project is to calibrate the vehicle odometry, thus reducing the need for manual tuning, and to automatically detect and avoid wheel slip for controlled, optimal racing.

I. INTRODUCTION

The F1Tenth simulator is idealized; it does not account for real-world imperfections in odometry, localization, and control. This leads to a sim-to-real gap, necessitating frequent tuning of software parameters through a manual, error-prone trial-and-error process.

Furthermore, a racetrack could consist of materials with various friction properties, and friction will decrease over time as tires accumulate dust. Without a detailed understanding of these materials, it is difficult to determine the braking, steering, and speed limits for a given track. Hence, operating the vehicle at a fixed friction coefficient is prone to failure.

Our method is composed of three components; an odometry tuner that calibrates the VESC steering and speed parameters against a ground-truth localization source, a slip detector that identifies and tracks waypoints at which wheel slip occurs, and an online optimizer that adjusts the velocity profile of the race line on subsequent laps to reduce slip. Together, these components form a control stack that maximizes vehicle speed while minimizing deviation from the intended race line.

II. RELATED WORK

A. Odometry Calibration

Odometry calibration is a well-studied topic with established techniques for a variety of robot and sensor configurations. One type of model fits odometry parameters through a regression minimizing the error between position measurements and an external source of ground truth, such as a predetermined trajectory [3]. Another model uses local deviations in the derivative of the system, comparing measurements with an idealized model, and fitting parameters over discrete time using a Kalman Filter [1]. Another technique, which can be applied generally, is collecting odometry measurement data on an arbitrary trajectory and then tuning odometry parameters using nonlinear optimization tools to fit the data [2]. However, calibration methods must be specialized to match a specific

robot model’s physical constraints, such as differential drive or Ackermann steering. Lee et. al. developed an extended Kalman filter approach with a fixed trajectory to calibrate a car-like vehicle [4]. This implementation can serve as a basis for models that use an arbitrary trajectory to make the calibration method easier to execute.

B. Slip Detection and Avoidance

Multiple methods have been proposed to detect and avoid wheel slip. In particular, reference [6] detects slip by comparing velocity and steering commands against measured velocity and heading estimated by fusing LiDAR localization and odometry data using an extended Kalman Filter. Its algorithm is implemented and tested on the F1Tenth platform. Reference [7] uses neural network to predict wheel slip ratio based on raw IMU data, bypassing the integration of IMU data into velocity estimation, to prevent the accumulated error of integration.

III. APPROACH

A. Odometry Tuner

The F1Tenth car’s wheel odometry is derived from the drive motor’s electrical rotation per minute (ERPM) as it drives the rear axle and the controlled response of the steering servo. Both signals are imperfect measurements because the wheels may slip and the steering servo has an inconsistent nonlinear response. The wheel odometry tuner seeks to minimize odometry errors so the car’s position between localization updates is as accurate as possible. This accuracy will reduce the influence of error compensation in the control loop, allowing for more reliable execution of the intended race line.

A physically-accurate digital twin of the lockers area in Tepper School of Business, Carnegie Mellon University, which we generally use for experimenting with the car, was created. The map resolution was defined as $1\text{ px} = 0.5\text{ cm}$, with a measurement accuracy of 1 cm.

Prior to running the calibration tests, we reconfigured the VESC firmware settings to ensure that the odometry inputs themselves were reliable. We tuned the VESC’s internal PID gains for accurate and stable ERPM control, enabled High Frequency Injection (HFI) to maintain accurate ERPM tracking even at low speeds where back-EMF sensing is unreliable, and calibrated the onboard IMU for accurate linear acceleration estimation. These adjustments produce consistent ERPM control across the vehicle’s full speed range and a reliable acceleration signal for downstream slip detection, ensuring the subsequent odometry tuning operates on trustworthy raw measurements rather than compensating for hardware-level noise.

The odometry tuner that we have implemented relies on a controlled scenario combined with an external source of ground truth. In order, we devised four tests to tune the following parameters: the steering angle to servo position offset, the steering angle to servo position gain, the speed to ERPM offset, and the speed to ERPM gain. The source of ground truth is the vehicle's pose as estimated by a particle filter localizing against the digital twin map described above. There is a human in the loop to change the odometry parameters using the results of each test. We chose this method due to its simplicity and reliability. There are few confounding variables when the tests are performed in the order that we specify.

Throughout each test, the odometry tuner sends continuous commands to drive the car at a fixed steering angle (δ) and fixed speed (v). Let l_d denote the dead-reckoning distance integrated from wheel encoder odometry and l the actual distance traveled, measured by the particle filter. The first two tests calibrate the steering pipeline by comparing the theoretical curvature predicted by the kinematic bicycle model against the measured curvature obtained by fitting a circle to the particle filter trajectory. We first tune the steering angle to servo position offset by commanding $\delta = 0$ and adjusting the offset until the car drives in a straight line. We then tune the steering angle to servo position gain by commanding a non-zero δ and adjusting the gain until the measured curvature matches the theoretical value $\kappa = \tan(\delta)/L$, where L is the wheelbase. The remaining two tests calibrate the speed pipeline by comparing the dead-reckoning distance against the actual distance. We tune the speed to ERPM offset by commanding $v = 0$ and adjusting the offset until $l_d = 0$, eliminating any residual creep. We then tune the speed to ERPM gain by commanding a non-zero v and adjusting the gain until $l_d = l$.

B. Online Slip Avoidance

We considered two strategies for detecting wheel slip during a race. The first compares the vehicle's actual position against the planned race line and flags a slip event when the distance to the nearest waypoint exceeds a threshold. The second compares the wheel-encoder velocity against the IMU-integrated velocity and flags a slip event when their difference exceeds a threshold. We adopted the second strategy because it isolates traction loss from race-line tracking error: a controller that simply tracks the race line poorly would trigger the first strategy even on a high-traction surface, whereas the velocity-comparison approach responds only to genuine wheel slip.

In the F1Tenth race, the track may consist of varying ground materials, such as carpet and concrete flooring. Modeling the vehicle dynamics using a fixed friction coefficient leads to imprecise control of the vehicle's maximum speed. Therefore, the goal of the online slip preventer is to detect wheel slip during the race and to reduce speed along the sections of the track where wheel slip occurred.

The online slip avoidance system monitors linear velocity error as "slip". To detect slip, we compare the wheel odometry velocity (based on how fast the wheels are rotating) to the

IMU-integrated velocity (based on how much the body of the vehicle accelerates). Because IMU measurements always have errors, IMU integration is tuned to balance sensitivity and accuracy by adjusting the amount of time during which IMU measurements are integrated: the duration D in seconds. At the start of each update tick, the IMU velocity is assumed to be the same as the wheel odometry velocity from D seconds ago. Then, Euler integration is used to obtain the vehicle velocity at the "present moment". A slip is detected when the linear velocity error exceeds a chosen threshold.

When slip is detected at waypoint (i), the speed limit at each waypoint (k) within a fixed radius is updated according to

$$\text{err_v} = \sqrt{\text{err_vx}^2 + \text{err_vy}^2} \quad (1)$$

$$\text{slip_ratio} = \min\left(\frac{\text{err_v}}{\text{slip_sensitivity}}, 1.0\right) \quad (2)$$

$$\alpha = 1.0 - \text{slip_ratio} \quad (3)$$

$$\text{new_speed} = \max(\text{min_speed}, \text{wp_speeds}[k] \cdot \alpha) \quad (4)$$

where e_{v_x} and e_{v_y} are the components of the velocity error between the wheel-encoder and IMU-integrated velocities, σ is a sensitivity parameter that controls how aggressively the speed is reduced for a given error magnitude, v_k is the previous speed limit at waypoint k , and $v_{\min}$ is a lower bound preventing the vehicle from being slowed below a usable speed.

After a few laps, this procedure converges on a trajectory within the vehicle's traction limits. The speed decrease is relative to the amount of deviation such that a larger deviation from ideal driving causes a greater slowdown in the next lap, leading to faster convergence.

IV. RESULTS

All videos of the performance of our car can be found at: <https://drive.google.com/drive/folders/1QQaqXFS-IDTpb4KJwo2quZHJLW8buzhM>. Videos are organized by Labs, Races, and UPenn race.

A. Odometry Tuner

There are two components of odometry to evaluate for the tuner:

- 1) Minimization of Linear Error
- 2) Minimization of Angular Error

Based on our proposal, our benchmark for success is achieving a linear odometry error of less than 4 cm and 0.37 radians of angular odometry error per meter traveled. These numbers were chosen based on the update rate of the sensors available to the car.

The accuracy of the odometry can be verified using an external source of ground truth for the car's position, like:

- 1) External motion capture system



Fig. 1. Dimensionally Accurate Map of Lockers in Tepper Basement

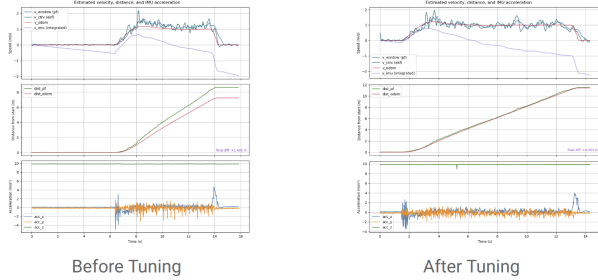


Fig. 2. Quantitative Results of Odometry Tuning

- 2) Localization within a dimensionally accurate map
- 3) SLAM using IMU and lidar only

We intended to use an external motion capture system because it provides high accuracy. However, the capture areas available to us were too small to cover the trajectories we needed for verification. Instead, we used physical measurements of a space to create a dimensionally accurate 2D with a precision of 1/2 cm, as shown in figure 1.

After completing the odometry tuning procedure with our odometry error measurement tools, we achieved an error of 0.45 cm and 0.006 radians per meter traveled, as shown in figure 2.

B. Online Slip Avoidance

To evaluate the effectiveness of the online slip avoidance system, the car will be commanded to follow a pre-defined path on a track with a low friction material. The slip avoidance system will be considered successful if it achieves at least 2 consecutive laps with no slipping.

Figure 3 displays all aspects of our system working together. The calibrated odometry ensures the system is much more responsive to input commands. The slip detector is reliably

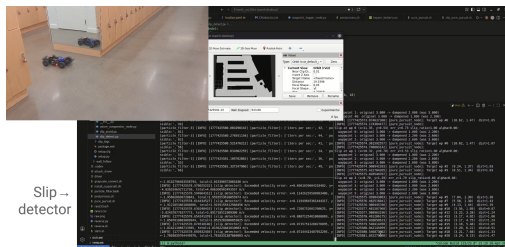


Fig. 3. Screenshot of Online Slip Avoidance controller

detecting slip events based on the IMU's velocity and wheel odometry's velocity. The controller has already applied a reduction of max speed at the closest node(s) which reduces the speed of the race car on the next lap. This ensures that the race car follows the desired raceline established by the raceline optimizer.

V. FUTURE WORK

The current odometry tuner currently relies on a human in the loop to adjust odometry variables in the Fltenth stack's VESC driver configuration. However, this process could be performed automatically with a simple feedback mechanism like a PI controller. The VESC driver must be modified to accept runtime changes to odometry parameters to this end.

In addition, the slip detector currently relies on proper tuning of several variables to have good performance. Specifically, the IMU integration duration D and the amount by which the speed is decreased must be tuned to desired values to meet the desired behavior of accurate slip detection, no/low slip, and a fast lap time. Furthermore, the slip detector could inform locations on the race line where there is headroom for higher speed, allowing for automatic, opportunistic lap time improvements.

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VI. APPENDIX

Our final contribution to Fltenth will be various ideas for confusing passwords (for WiFi, SSH, etc.) to ensure rogue teams do not connect to your vehicle maliciously, because "teamX" is not very secure. Imagine the following being the response to the question "what is the password for your Jetson?" or "What is your WiFi password?":

- 1) all lowercasewithspace
- 2) allLowerCaseInCamelCase
- 3) onthebackoftherouter
- 4) sudo rm -rf --no-preserve-root /*
- 5) thefirstninedigitsofpi
- 6) JimiDhon123
- 7) @grok generate a password
- 8) Where did you get those coconuts?
- 9) We found them.
- 10) You found them? This is a temperate zone, coconuts are tropical.
- 11) Swallows fly south for the winter, but these are not strangers to our lands.
- 12) Are you suggesting coconuts migrate?
- 13) Not at all, they could have been carried.
- 14) You know, the password.
- 15) The password for Kuzco.
- 16) The password specifically meant for Kuzco.
- 17) Kuzco's password.
- 18) oneTwoTHREE4
- 19) According to all known laws of aviation, there is no way a bee should be able to fly. Its wings are too small to get its fat little body off the ground. The bee, of course, flies anyway because bees don't care what humans think is impossible. Yellow, black. Yellow, black. Yellow, black. Yellow, black. Ooh, black and yellow! Let's shake it up a little. Barry! Breakfast is ready! Coming! Hang on a second. Hello? Barry? Adam? Can you believe this is happening? I can't. I'll pick you up. Looking sharp. Use the stairs, Your father paid good money for those. Sorry. I'm excited. Here's the graduate. We're very proud of you, son. A perfect report card, all B's. Very proud. Ma! I got a thing going here. You got lint on your fuzz. Ow! That's me! Wave to us! We'll be in row 118,000. Bye! Barry, I told you, stop flying in the house! Hey, Adam. Hey, Barry. Is that fuzz gel? A little. Special

day, graduation. Never thought I'd make it. Three days grade school, three days high school. Those were awkward. Three days college. I'm glad I took a day and hitchhiked around The Hive. You did come back different. Hi, Barry. Artie, growing a mustache? Looks good. Hear about Frankie? Yeah. You going to the funeral? No, I'm not going. Everybody knows, sting someone, you die. Don't waste it on a squirrel. Such a hothead. I guess he could have just gotten out of the way. I love this incorporating an amusement park into our day. That's why we don't need vacations. Boy, quite a bit of pomp under the circumstances. Well, Adam, today we are men. We are! Bee-men. Amen! Hallelujah! Students, faculty, distinguished bees, please welcome Dean Buzzwell. Welcome, New Hive City graduating class of 9:15. That concludes our ceremonies And begins your career at Honex Industries! Will we pick our job today? I heard it's just orientation. Heads up! Here we go. Keep your hands and antennas inside the tram at all times. Wonder what it'll be like? A little scary. Welcome to Honex, a division of Honesco and a part of the Hexagon Group. This is it! Wow. Wow. We know that you, as a bee, have worked your whole life to get to the point where you can work for your whole life. Honey begins when our valiant Pollen Jocks bring the nectar to The Hive. Our top-secret formula is automatically color-corrected, scent-adjusted and bubble-contoured into this soothing sweet syrup with its distinctive golden glow you know as... Honey! That girl was hot. She's my cousin! She is? Yes, we're all cousins. Right. You're right. At Honex, we constantly strive to improve every aspect of bee existence. These bees are stress-testing a new helmet technology. What do you think he makes? Not enough. Here we have our latest advancement, the Krelman. What does that do? Catches that little strand of honey that hangs after you pour it. Saves us millions.